

- 5 (a) work done per / on unit positive charge
 moving charge from infinity to the point
- (b) (i) α -particle and gold nucleus repel each other B1
all kinetic energy of α -particle converted into electric potential energy B1 [2]
- (ii) 1 potential energy = $(79 \times 2 \times \{1.6 \times 10^{-19}\}^2) / (4\pi \times 8.85 \times 10^{-12} \times d)$ C1
 kinetic energy = $4.8 \times 1.6 \times 10^{-13} = 7.68 \times 10^{-13} \text{ J}$ C1
 equating to give $d = 4.7 \times 10^{-14} \text{ m}$ A1 [3]
- (ii) 2 $F = Qq / 4\pi\epsilon_0 d \times 1 / d = 7.68 \times 10^{-13} \times 1 / (4.7 \times 10^{-14})$ C1
 = 16 N A1 [2]

[Total: 9]

- 6 (a) concentric circles (at least three lines)

M1